

[DC] Effects of Asymmetric Locomotion Methods on Collaborative Navigation and Wayfinding in Shared Virtual Environments

Soumyajit Chakraborty *
Vanderbilt University, USA

ABSTRACT

Navigation and wayfinding can be accomplished either by single person or a group of people. Using the help of immersive virtual reality technology, significant research has been conducted to find out how a person can navigate and wayfind in a virtual world. However, there has been little work done that asks how multiple people can collaboratively navigate and wayfind in a virtual world. In this proposal, we investigate this question with a specific interest on how different locomotion methods can affect the acquired knowledge of a group of individuals in a distributed, shared virtual environment.

Keywords: Virtual reality, Locomotion methods, Collaborative navigation, Distributed virtual environments

1 INTRODUCTION

Distributed, shared virtual reality (VR) systems allow multiple users to participate in collaborative activities. A common collaborative activity in the real world is joint exploration, for example where people tour a city or park together. Shared virtual environments offer the same opportunities. However, little study has been done of how to collaboratively explore and navigate in shared virtual environments.

This proposal explores the design of collaborative systems for navigating and wayfinding in shared virtual environments. The focus is on locomotion methods. In particular, we focus on how users employing different locomotion methods can effectively navigate and wayfind as a dyad or larger group. We refer to this as the *asymmetric locomotion interface problem*. Specifically, the question we ask is: How well can users collaboratively navigate and wayfind when they are employing different locomotion methods?

We start by studying dyads and then extend to larger groups. Dyads have been studied before by Buck et al. [1]. They showed that dyads using a symmetric locomotion method can acquire better spatial knowledge than individuals in an immersive virtual environment. Weissker et al. formulated a comprehensible group navigation technique in which the guide of a group can teleport all the group members from one place to another in a virtual museum [10]. Similarly, Dong et al. created a technique for multiple users in a shared virtual environment. This technique will not only keep users from bumping into the physical boundary of the virtual world, but also it will detect the possible incoming collision with another roaming user in the virtual world, and redirect both of them towards safe collision free directions [3]. All the previous research has been conducted using symmetric locomotion methods, i.e., all the participants were using the same locomotion method while navigating in the virtual world.

As the prior work demonstrates there are many ways to assess success in collaborative navigation. Habgood et al. compared three different methods of locomotion techniques while assessing presence and simulator sickness in the virtual world [5]. On the other hand,

*e-mail: soumyajit.chakraborty@vanderbilt.edu



Figure 1: A top down view of the maze presented to the individuals (left) and the dyads (right) following that of Buck et al. [1]

spatial awareness and memory were assessed by Coomer et al. by comparing four different locomotion methods [2]. Four different video game locomotion techniques were compared by Paris et al. to assess how simulator sickness and self-motion cues of the users are affected by these locomotion interfaces while navigating in a virtual world [8]. In this proposal we focus on spatial learning, specifically we intend to ask how well groups can acquire survey knowledge of their environment. The prior work was conducted using individual navigators in a virtual world. How different locomotion methods used by multiple users in a shared virtual environment will interact is unknown.

Clearly, there are a large number of possible locomotion methods to choose from when navigating in a virtual environment. In this proposal we will only select a few, representative methods to study. In particular, we will study *joystick with body-based rotations* [9], *resetting* [7], and *walking in place* [6]. These methods are widely used to explore large virtual environments as well as to gain spatial learning. We chose these because the joystick based locomotion method is an example of continuous steering method. People can physically walk in the tracked space in the case of resetting, walking in place is an examples of a hybrid locomotion method.

2 RESEARCH METHODOLOGY

This research proposal will try to investigate the answers of three main research questions. First we will understand the effects of two asymmetric locomotion methods on dyads. Then, we will see the effect of asymmetric locomotion methods on multiple distributed individuals in a shared virtual environment.

As described above, our primary performance metric by which to discriminate locomotion methods will be how well a user is able to acquire survey knowledge using them. To determine this we employ an experimental paradigm similar to that of Buck et al. [1] in which participants learned the layout of objects in a maze, and are tested on this knowledge. In Buck et al., they learned individually and as part of a dyad. An example of the individual and dyadic maze layouts we might use is shown in Figure 1.

RQ1: In the presence of two asymmetric locomotion methods, how well do dyads acquire survey knowledge

in a shared virtual environment? We start our investigation between physically distributed dyads along with two asymmetric locomotion methods such as *joystick with body-based rotations* and *resetting*. To check the effect of the locomotion methods, we will set up an experiment with three conditions on the dyad: (i) both members of the dyad use joystick (J) locomotion; (ii) both members use resetting (R); and (iii) one member uses joystick locomotion and one member uses resetting. Prior work has shown $J < R$ in terms of acquiring survey knowledge. Our hypotheses are as follows:

- **H1.1:** (i) $JJ < RR$ and (ii) $JJ < JR < RR$
- **H1.2:** (i) $J < JJ$ and (ii) $R < RR$
- **H1.3:** (i) $J < JR$ and (ii) $R < RJ$

We choose joystick locomotion to start with because it was the method used by Buck et al. [1]. We choose to compare it to resetting because resetting is as close as we can get to real walking in our laboratory.

RQ2: Given that a certain locomotion method < another asymmetric locomotion method, does it still hold when dyads use these methods while acquiring survey knowledge in a shared virtual environment? Investigating this research question, we continue to focus on dyads in this research question to understand how spatial knowledge is affected by different locomotion methods as we move from a single individual to a collaborative activity. We add walking in place in this research question. We will repeat the experiments of RQ1 except we will compare joystick locomotion to walking in place, and walking in place to resetting. Our hypothesis is that if a locomotion method is < another locomotion method for single individuals, then the same relations will hold as were found in RQ1. Our additional hypothesis is that if one locomotion method is equivalent to another method in terms of spatial knowledge acquisition for a single user, then those methods will be equivalent for the dyad. We make this additional hypothesis because some prior work has shown that walking in place and resetting are equivalent [7].

RQ3: Will the previous findings still hold if the group size increases while the group members acquire survey knowledge in a shared virtual environment? If the group size increases beyond two, group navigation behavior will become more difficult to predict. Previous research has shown that a group member can emerge as a leader of a group which affects the group dynamics [4]. We propose to run similar experiments to the previous sets of experiments with groups of three members. We have two groups of participants, those in a symmetric locomotion condition where we fix the locomotion method over the entire experiment, e.g., joystick locomotion. The second group of three participants will be the asymmetric group, with one member having the resetting method, one having a walking in place method, and one having joystick locomotion. The task is the same as before where we measure individual spatial knowledge acquisition and then compare it to group acquisition. We conjecture that a leader will emerge from the triad; we propose to determine the leader from analysis of audio transcripts of the sessions. We hypothesize that spatial knowledge acquisition will be determined by the leader. That is, if the leader uses an inferior locomotion method for spatial knowledge acquisition then the group spatial knowledge acquisition will suffer. A third hypothesis involves the relation of the symmetric group to the asymmetric group. This hypothesis is difficult to formulate without knowing the results of RQ1 and RQ2 but we conjecture that the results of RQ2 will generalize to a larger group.

Because of inadequate prior research in the related field, it is hard to expect the results accurately and compare the findings with the previous results. So, this finding is crucial, as it will provide a good understanding of the effect of locomotion methods, and group dynamics on acquired survey knowledge by a group in a shared virtual environment, which will help future virtual reality application developers to understand the effect of various locomotion methods

in a shared virtual environment and take necessary steps to build an immersive experience for all the attending participants in that environment.

3 CONCLUSIONS

To this date, a little is known about collaborative navigation and wayfinding in a shared virtual environment with multiple distributed individuals. This work will add a significant amount of knowledge to the available literature. Understanding how different locomotion modes work together in shared virtual environments is important as this type of interaction can be expected to become more common as different hardware platforms proliferate.

4 OPEN QUESTIONS

1. The feasibility of this proposal requires restricting the number of locomotion methods to a small number. Have we chosen the best set of locomotion methods to test? How can we think about generalizing our results to other locomotion methods without directly testing them?
2. The feasibility of the experiment for RQ3 seems to require testing only a small group. Is there a way to test larger group of people and keep it manageable?
3. In RQ3, we assume that a leader will emerge. Is that a good assumption? Is audio transcript analysis the best way to determine the leader?

REFERENCES

- [1] L. E. Buck, T. P. McNamara, and B. Bodenheimer. Dyadic acquisition of survey knowledge in a shared virtual environment. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 579–587. IEEE, 2020.
- [2] N. Coomer, S. Bullard, W. Clinton, and B. Williams-Sanders. Evaluating the effects of four vr locomotion methods: joystick, arm-cycling, point-tugging, and teleporting. In *Proceedings of the 15th ACM symposium on applied perception*, pp. 1–8, 2018.
- [3] T. Dong, Y. Shen, T. Gao, and J. Fan. Dynamic density-based redirected walking towards multi-user virtual environments. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, pp. 626–634. IEEE, 2021.
- [4] J. J. Faria, J. R. Dyer, C. R. Tosh, and J. Krause. Leadership and social information use in human crowds. *Animal Behaviour*, 79(4):895–901, 2010.
- [5] M. J. Habgood, D. Moore, D. Wilson, and S. Alapont. Rapid, continuous movement between nodes as an accessible virtual reality locomotion technique. In *2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 371–378. IEEE, 2018.
- [6] S. Hanson, R. A. Paris, H. A. Adams, and B. Bodenheimer. Improving walking in place methods with individualization and deep networks. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 367–376. IEEE, 2019.
- [7] R. Paris, M. V. Joshi, Q. He, G. Narasimham, T. P. McNamara, and B. Bodenheimer. Acquisition of survey knowledge using walking in place and resetting methods in immersive virtual environments. In *Proceedings of the ACM Symposium on Applied Perception, SAP '17*. ACM, New York, NY, USA, 2017.
- [8] R. Paris, J. Klag, P. Rajan, L. Buck, T. P. McNamara, and B. Bodenheimer. How video game locomotion methods affect navigation in virtual environments. In *ACM Symposium on Applied Perception 2019*, pp. 1–7, 2019.
- [9] B. E. Riecke, B. Bodenheimer, T. P. McNamara, B. Williams, P. Peng, and D. Feuerissen. Do we need to walk for effective virtual reality navigation? physical rotations alone may suffice. In *Spatial Cognition*, pp. 234–247, 2010.
- [10] T. Weissker and B. Froehlich. Group navigation for guided tours in distributed virtual environments. *IEEE Transactions on Visualization and Computer Graphics*, 27(5):2524–2534, 2021.